Project Proposal for Deep Learning CS 7643

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1 Team Name

YikesDetector

2 Project Title

Toxic Comment Classification

3 Project Summary

People are spending more and more time online; a typical internet user spends 2.5 hours per day on social media [4]. Individuals interact with each other or with an audience through various platforms and messaging services, including Instagram, Facebook, TikTok, Reddit, Twitter, and YouTube.

Although these platforms offer many benefits, they also serve as ubiquitous vectors for bullying and hate, especially among young people. Studies have shown a rising rate of adolescent depression in the United States [12], and individuals receiving negative feedback on social media posts are more susceptible to emotional distress [7]. The volume of toxicity on social networks has also increased since the COVID-19 pandemic [8]. "Evidence suggests that the private incentives of social media platforms to reduce toxicity may not align with social needs" [1].

To help address this problem, we aim to reliably detect toxicity in comments on social media as the first step toward prevention is detection. This topic interests us given the growth and severity of the problem and its potential impact.

4 Approach

We will leverage a dataset maintained by Google, which contains a large number of Wikipedia comments (see Section 6), to train various model architectures.

These comments have been labeled by human reviewers for toxic behavior. The types of toxicity include:

- Toxic
- Severe toxic
- Obscene
- Threat
- Insult
- Identity hate

We plan to train several models for this classification task: a full Transformer encoder-decoder, BERT, and Llama with fine-tuning. A subset of this work will involve replicating and validating an existing paper on this topic (see Section 5). We will compare the accuracy of these models in detecting toxicity and also run an LLM classification based on the paper [11] to see how it performs compared to our tuned models.

Finally, we will download comments from a social media platform to test our model's performance on real data and conduct a qualitative analysis. If possible, comparing overall toxicity rates across platforms would be interesting.

As a stretch goal, we want to explore possible directions to improve the related papers. One potential path is to find additional data sources to use for training and testing, as Wikipedia-style comment may not carry over well to other platforms. One possible approach is to use a combination of other small datasets [6].

As a secondary stretch goal, we can conduct an explainability analysis to investigate the relationship between the text and the model's classification decisions. This aligns with the approach used in the paper [3]. This analysis will help us understand the model's behavior before deploying it in a real-world setting.

5 Resources

Recent research in toxic comment classification strongly supports the use of transformer-based architectures, including BERT, RoBERTa, and DistilBERT, as these consistently outperform traditional models such as CNNs and LSTMs. The key advantage of transformers lies in their powerful contextual embeddings, which capture nuanced language patterns and subtle forms of toxicity ([5][10]. Ref [9]) specifically highlighted BERT's ability to significantly improve performance on toxicity detection tasks because of its deep understanding of context within comments. Furthermore, ref. [2] emphasized the critical need to address biases associated with sensitive identity terms, advocating for effective balanced data handling strategies to ensure fairness.

6 Datasets

The dataset is taken from the Toxic Comment Classification Challenge on Kaggle and is now maintained by Google. It can be accessed at:

- https://huggingface.co/datasets/google/jigsaw_toxicity_pred

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